

Redistricting and Democratic Outcomes: A Framework

Track O Overview: Electoral Competitiveness, Voter Turnout, Legislative Polarization, and Constituent-Representative Distance

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Abstract

Congressional redistricting determines not only who represents each community but shapes the democratic health of the legislature itself. Yet most algorithmic redistricting research evaluates maps on geometric properties—compactness, population equality, minority representation—without connecting these properties to the downstream democratic outcomes that redistricting is ultimately meant to serve. This paper establishes a framework linking map geometry to four downstream outcome dimensions: electoral competitiveness (the frequency and margin of competitive elections), voter turnout (participation rates conditional on district competitiveness), legislative polarization (the ideological distance between parties’ median members), and constituent-representative geographic distance (mean travel time from constituent to representative office).

We survey prior literature across all four dimensions, with particular attention to the quasi-experimental identification strategies that exploit redistricting cycles to establish causal rather than merely correlational relationships. Key priors: Caughey and Tausanovitch [2017] show that state legislative maps affect policy outcomes through ideological selection; Rodden [2019] demonstrates that geographic sorting produces a structural Democratic efficiency disadvantage that gerrymandering amplifies; Jacobson [2015] documents the secular decline of competitive seats from the 1990s through the 2020 cycle.

The central prediction of the O-series, developed formally in O.1–O.4, is that compact algorithmic plans produced by BISECT predict better outcomes on all four dimensions independent of partisan composition. The mechanism: gerrymandering requires non-compact geometry to achieve its effects (packing and cracking create district

“necks” that increase travel distance, reduce intra-district political heterogeneity, reduce competitiveness, and select for ideologically extreme candidates). Compact plans disrupt these mechanisms even when they are not drawn to produce any particular partisan outcome.

This framework paper synthesizes the O.1–O.4 companion papers and provides the identification strategy, theoretical foundations, and measurement protocols used throughout the O-series.

Keywords

Redistricting, democratic outcomes, electoral competitiveness, voter turnout, legislative polarization, constituent-representative distance, quasi-experiment, algorithmic redistricting, BISECT, gerrymandering

1 Introduction

Congressional redistricting is, at bottom, an exercise in democratic engineering. Mapmakers decide—whether deliberately or through algorithmic neutrality—which communities share a representative, which elections will be competitive, which incumbents face meaningful challenges, and how far constituents must travel to access their representative’s office. The academic redistricting literature has devoted substantial effort to measuring the partisan properties of maps (the L-series), evaluating their compactness (the K-series), and analyzing their VRA compliance (the D-series). What it has devoted less effort to is systematically linking geometric map properties to the downstream democratic outcomes that redistricting is constitutionally required to serve.

This gap matters practically. Courts evaluating redistricting plans ask not just whether a map is partisan but whether it harms democratic representation. State constitutional provisions guaranteeing “free and equal elections” (Pennsylvania), “fair districts” (Michigan), or requiring that districts “not dilute” minority voting strength (Voting Rights Act) are ultimately about outcomes, not geometry. The geometric metrics we can compute from a shapefile are useful only insofar as they predict the outcomes that the constitutional provisions care about. If compactness predicts electoral competitiveness and travel distance but not legislative polarization, then compactness arguments in court are appropriately limited.

The O-series papers (O.1 through O.4) investigate four outcome dimensions, each in dedicated empirical analysis. This overview paper provides the integrating framework:

- **O.1 — Electoral Competitiveness:** The distribution of competitive seats under

enacted vs. BISECT plans, using VEST precinct-level data and a within-state difference-in-differences identification across redistricting cycles.

- **O.2 — Voter Turnout:** Precinct-level turnout as a function of district competitiveness, instrumented by the redistricting-induced variation in predicted margin.
- **O.3 — Legislative Polarization:** DW-NOMINATE score gaps between the two parties’ congressional medians, using redistricting-cycle reshuffling as a natural experiment.
- **O.4 — Constituent-Representative Distance:** Mean driving time from census tract centroid to the representative’s district office address, computed via OSRM routing API, with Polsby-Popper as the predictor.

The organizing prediction across all four papers is stated at the end of this paper (Section 5): *compact algorithmic plans predict better outcomes on all four dimensions, independent of partisan composition*. This prediction is testable, falsifiable, and has policy implications that extend beyond the typical gerrymandering debate. Even in a world without partisan gerrymandering, the choice between compact and non-compact maps would matter for democratic quality.

1.1 Why Outcomes, Not Just Geometry

The academic literature on gerrymandering has generated a rich set of geometric measures (Section 4), but geometric measures are not self-justifying. A court that invalidates a map because its Polsby-Popper score is 0.12 is making an implicit empirical claim: that Polsby-Popper scores of 0.12 are associated with democratic harms that the constitution prohibits. Without the empirical link from geometry to outcomes, the constitutional argument is incomplete. Courts have sometimes rejected compactness arguments precisely because the link was asserted but not demonstrated [vei, 2004].

The BISECT platform provides a natural test bed for establishing this link. Because BISECT maps are generated without partisan data, any differences in democratic outcomes between BISECT plans and enacted plans that survive controls for partisan geography must be attributable to geometric features of the maps themselves. This identification logic—partisan geography controlled, geometry varied—is the backbone of the O-series.

1.2 The 2020 Redistricting Cycle as Primary Dataset

The O-series uses the 2020 redistricting cycle as its primary dataset because it is the first cycle for which all of the following are simultaneously available: (1) BISECT algorithmic plans for all 50 states; (2) VEST 2020 precinct-level election returns; (3) ACS 2019–2021 5-year estimates for tract-level demographic covariates; (4) LODES commuting data for travel distance validation; and (5) enacted plans that can be compared to algorithmic alternatives using a controlled before-after design.

The 2010 cycle serves as the pre-period for difference-in-differences estimates in O.1 and O.2. The 2000 cycle is used for robustness checks where VEST data coverage allows.

2 Four Outcome Dimensions: Definitions and Theoretical Foundations

The four outcome dimensions of the O-series are chosen to span the major empirical literature on redistricting and democratic representation. Table 1 summarizes each dimension’s definition, measurement unit, data source, and the O-series paper that provides the primary analysis.

Table 1: Four Outcome Dimensions: O-Series Framework

Dimension	Outcome Variable	Unit	Data Source	Paper
Competitiveness	Margin < 10% seats	Fraction of seats	VEST 2020	O.1
Turnout	Precinct VAP turnout	Percentage points	VEST 2020	O.2
Polarization	DW-NOMINATE gap	Score units	VoteView	O.3
Const.-Rep. Distance	Mean driving time	Minutes	OSRM/OSM	O.4

2.1 Dimension 1: Electoral Competitiveness

A district is *competitive* if the two-party vote margin in a given election falls within 10 percentage points of the 50% threshold. We use the two-party presidential vote share as the competitiveness predictor rather than the congressional vote share, because presidential vote is more stable across elections and less subject to incumbency effects. The fraction of competitive seats in a state’s congressional delegation is the aggregate competitiveness measure.

Partisan gerrymandering systematically reduces competitiveness by design: the optimal partisan map packs opposition voters into a small number of safe seats while cracking re-

maining opposition voters into a larger number of seats where they lose by modest margins. Both moves reduce competitive seats. Compactness disrupts this: compact districts tend to include geographically proximate communities that are more politically similar, creating more homogeneous districts that are either safe or marginal—but the marginal districts are not chosen by the mapmaker.

The theoretical prediction for O.1 is that BISECT plans produce more competitive seats on average relative to enacted gerrymanders in states where the enacted map has a documented partisan slant ($|EG| > 0.08$), but that the direction of the competitiveness difference is unpredictable in states without documented partisan gerrymandering. This asymmetric prediction is important: compact plans are not maximally competitive. They are competitive-neutral, which means they produce more competitive seats than gerrymandered plans but not necessarily more than the maximum achievable under algorithmic competitive maximization.

2.2 Dimension 2: Voter Turnout

Voter turnout at the precinct level is a function of multiple factors: mobilization, registration accessibility, partisan intensity, candidate quality, and—the channel of interest here—district competitiveness. The “stimulation hypothesis” in electoral participation research [Hays, 1964] posits that competitive elections increase turnout by raising the perceived electoral stakes for individual voters. If redistricting affects competitiveness (O.1), it may affect turnout through this channel.

The identification challenge for O.2 is separating the redistricting-induced competitiveness variation from the confounding of partisan geography. A precinct moved from a safe district into a competitive district may show increased turnout not because of the redistricting change per se but because competitive districts attract more campaign resources and candidate quality. We address this using a shift-share instrument: the precinct-level predicted margin under the new map, constructed from the precinct’s presidential vote history and the new district boundaries, serves as the instrument for realized competitiveness.

2.3 Dimension 3: Legislative Polarization

Legislative polarization—the ideological distance between the parties’ congressional medians—has increased monotonically since the 1970s [McCarty et al., 2006]. The redistricting literature’s claim is that partisan gerrymandering contributes to polarization by reducing competitive seats, which in turn selects for more ideologically extreme candidates in safe-seat primaries [Fiorina and Abrams, 2008]. The mechanism has two steps: (1) fewer competitive seats $\rightarrow$ more safe seats $\rightarrow$ more primary-determined general elections; (2) primary elec-

torates are more ideologically extreme than general electorates $\rightarrow$ extreme candidates win safe-seat primaries.

The empirical evidence for both steps is weaker than the theory suggests. Caughey and Tausanovitch [2017] find that redistricting affects state legislative outcomes through ideological selection in some contexts but not universally. McCarty et al. [2009] find that polarization has increased even in states without gerrymandering, suggesting geographic sorting as an independent driver. The O.3 analysis uses a DW-NOMINATE based approach with state-level controls for geographic sorting to isolate the redistricting contribution.

2.4 Dimension 4: Constituent-Representative Geographic Distance

Geographic distance between constituents and their representative is a direct measure of representational access that has received surprisingly little empirical attention. The intuition is straightforward: a constituent living in a district drawn as a narrow corridor connecting two distant communities faces a longer trip to the representative’s local district office than a constituent in a geographically compact district. This matters because district office access is the primary channel through which constituents raise constituent services issues (visa applications, Social Security disputes, veterans’ benefits, and similar casework).

We operationalize this as mean driving time from each census tract’s population-weighted centroid to the representative’s primary district office address, as listed on house.gov. OSRM (Open Source Routing Machine) with OpenStreetMap road network data provides the routing computation. The O.4 analysis, the most novel in the O-series, documents a correlation of $r = -0.67$ between district Polsby-Popper compactness and mean constituent driving time across the 435 congressional districts, and shows that BISECT plans reduce mean driving time from 48 minutes (enacted gerrymanders) to 34 minutes.

3 Identification Strategy: Redistricting Cycles as Quasi-Experiments

The central identification challenge for all four O-series papers is establishing causal rather than correlational relationships between map geometry and democratic outcomes. This section describes the identification strategies used across the series, with particular emphasis on the features that make redistricting cycles well-suited as quasi-experiments.

3.1 Redistricting Cycles as Natural Experiments

The decennial redistricting cycle generates a near-ideal natural experiment for studying the effects of map geometry on democratic outcomes. The experiment has three favorable properties:

Sharp discontinuity in treatment. District boundaries change abruptly following redistricting, creating a sharp discontinuity in treatment assignment (which district a precinct belongs to) at a known calendar date. Precincts whose boundaries are unchanged across the redistricting cycle serve as a natural comparison group, while precincts that change district assignment receive the treatment. This design is related to geographic regression discontinuity [Keele and Titiunik, 2015] but exploits temporal rather than spatial variation.

Pre-redistricting baseline. Outcomes from the two election cycles immediately before redistricting (typically four House elections spanning eight years) provide a pre-period baseline against which post-redistricting outcomes can be compared. Because redistricting occurs at the start of each decade, a full two-term congressional cycle is available in both the pre- and post-period under the same map.

Plausible exogeneity of assignment mechanism. While the decision to gerrymander is endogenous, the specific boundary placement within a gerrymandered map is driven by complex political calculations that include geographic constraints, VRA requirements, incumbent protection, and intra-party negotiation. This complexity makes the specific precinct-level district assignment difficult to fully anticipate from precinct-level outcome data, supporting an approximate exogeneity assumption for precinct-level analysis (distinct from the plan-level analysis, where endogeneity of gerrymandering is severe).

3.2 Difference-in-Differences Design

The primary identification strategy for O.1 and O.2 is a difference-in-differences (DiD) design that compares changes in outcomes for precincts that experienced a large district geometry change to precincts that experienced a small change, before and after redistricting.

Let Y_{pt} be the outcome (turnout, competitiveness indicator) for precinct p in election year t . Let ΔPP_p be the change in the Polsby-Popper score of the district containing precinct p between the pre- and post-redistricting cycles. The DiD estimator is:

$$Y_{pt} = \alpha + \beta \cdot \text{Post}_t + \gamma \cdot \Delta PP_p + \delta \cdot (\text{Post}_t \times \Delta PP_p) + \mathbf{X}'_{pt}\theta + \varepsilon_{pt} \quad (1)$$

where $\text{Post}_t = 1$ for election years after redistricting, $\mathbf{X}_{pt}$ is a vector of precinct-level covariates (median income, racial composition, urban-rural classification), and δ is the parameter of interest: the effect of a one-unit increase in Polsby-Popper change on the outcome

change following redistricting.

Standard errors are clustered at the district level to account for within-district correlation in outcomes, and at the state-year level to account for state-specific electoral shocks.

3.3 Instrumental Variables: The Algorithmic Instrument

For outcomes where the DiD design is subject to selection concerns (e.g., if states that gerrymander more aggressively also have systematically different political cultures that affect turnout independently of redistricting), we supplement the DiD with an instrumental variables design.

The instrument is the *algorithmic counterfactual geometry*: the Polsby-Popper score of the BISECT district that contains each precinct in the algorithmic plan, net of the enacted district Polsby-Popper. This instrument satisfies the relevance condition by construction (algorithmic geometry is highly correlated with actual geometry in states without gerrymandering, and systematically higher in gerrymandered states) and the exclusion restriction by design (the algorithmic plan is computed without reference to any outcome variable; partisan and turnout data play no role in BISECT’s partitioning).

The IV estimand recovers a local average treatment effect (LATE) for the subpopulation of precincts whose district assignment is affected by the compactness difference between the enacted and algorithmic plans—in practice, precincts near the district boundaries where gerrymandering produces the most non-compact geometry.

3.4 Event Study Design for Polarization (O.3)

For legislative polarization, the DiD design must be applied at the legislator level rather than the precinct level. The design uses redistricting-cycle transitions as event dates: for each member of Congress, we record their DW-NOMINATE score in each term and interact this with whether they entered the House after a redistricting cycle that changed their district’s geometry substantially.

The event study specification:

$$\text{DW-NOMINATE}_{it} = \alpha_i + \alpha_t + \sum_{\tau=-4}^4 \beta_{\tau} \cdot \mathbb{I}[\text{cycle}_t = \tau] \cdot \Delta\text{PP}_i + \varepsilon_{it} \quad (2)$$

where α_i is a member fixed effect, α_t is a Congress fixed effect, τ indexes election cycles relative to the redistricting event, and ΔPP_i is the Polsby-Popper change for member i ’s district in the redistricting event. Coefficients β_{τ} for $\tau < 0$ serve as pre-trend tests; β_{τ} for $\tau \geq 0$ estimate the dynamic treatment effect of compactness changes on polarization.

4 Prior Literature and Empirical Benchmarks

The O-series builds on three distinct strands of the redistricting literature. This section reviews each strand’s key findings and identifies the gaps that the O-series addresses.

4.1 Caughey, Tausanovitch, and State Legislative Effects

Caughey and Tausanovitch [2017] provide perhaps the most rigorous causal estimate of redistricting effects on policy outcomes. Using a regression discontinuity design that exploits close elections at the district margin, they show that the partisanship of a state legislature—determined in part by its district map—has significant causal effects on policy liberalism across a broad range of policy areas (taxes, spending, social policy). A one-standard-deviation increase in Democratic seat share produces a policy liberalism shift of roughly 0.4 standard deviations on their composite policy index.

For the O-series, the Caughey-Tausanovitch finding provides the motivation for the legislative polarization analysis (O.3): if partisan seat share affects policy, and redistricting affects partisan seat share, then redistricting affects policy through an ideological selection mechanism. The O.3 paper asks whether this mechanism operates through polarization specifically—whether compact plans produce legislators closer to the center, not just different partisan distributions.

The limitation of Caughey and Tausanovitch for the O-series is that their analysis is at the state legislative, not congressional, level. Congressional redistricting follows different legal constraints (one-person-one-vote with near-zero tolerance vs. the $\pm 5\%$ tolerance permitted in state legislative maps). Congressional districts are also larger, reducing the intra-district political heterogeneity that drives the competitive mechanisms in O.1 and O.2.

4.2 Rodden and Geographic Sorting

Rodden [2019] documents a structural feature of American political geography that is directly relevant to redistricting outcomes: Democratic voters are systematically more geographically concentrated (in cities and dense suburbs) than Republican voters. This geographic sorting creates a structural efficiency disadvantage for Democrats under any single-member district system: Democratic voters “waste” more votes in landslide districts than Republicans, and this waste is amplified by partisan gerrymandering.

The Rodden framework is important for the O-series in two respects. First, it clarifies the distinction between geography-induced inefficiency and gerrymandering-induced inefficiency. Any compact algorithmic plan will inherit some of the Democratic efficiency disadvantage

simply because it must cut between geographically sorted communities. The O.1 analysis must therefore control for the geographic sorting baseline when measuring competitiveness effects. Second, Rodden’s analysis implies that the constituent-representative distance effect (O.4) has a partisan asymmetry: Democratic constituents in urban districts will be closer to their representative regardless of gerrymandering, because urban representatives locate their offices in the city center that both the representative and constituents share.

Chen and Rodden [2013] extend the Rodden framework with a simulation-based approach, generating thousands of random redistricting plans for specific states and showing that partisan outcomes consistent with Republican advantage arise under most random plans in states like Florida and North Carolina—not from deliberate gerrymandering but from geographic sorting. The O-series must control for this baseline to isolate the gerrymandering contribution to democratic outcomes.

4.3 Jacobson and the Decline of Competitive Elections

Jacobson [2015] documents the secular decline of competitive House elections from 1946 to 2014. The fraction of House seats decided by margins under 10 percentage points declined from roughly 40% in the 1970s to under 20% by 2012. Jacobson attributes this decline to multiple factors: geographic sorting (consistent with Rodden), incumbency advantage, partisan polarization that reduces ticket-splitting, and redistricting.

For the O.1 analysis, Jacobson’s secular trend presents a confounding challenge: any redistricting effect on competitiveness must be separated from the secular decline. We address this by including state-specific time trends in the DiD specification and by focusing the analysis on redistricting-induced *changes* in competitiveness rather than levels. States where the 2020 redistricting cycle produced maps with substantially higher Polsby-Popper scores than the 2010 cycle (naturally or algorithmically) are the treatment group; states with stable geometry are the control group.

4.4 Additional Literature

Turnout and competitiveness: Cox and Munger [1989] provide a comprehensive review of the mobilization effects of electoral competition, finding consistent positive effects across electoral systems. Abramowitz [2010] document the decline of swing voters and its implications for competitive-election mobilization effects. The O.2 analysis uses these as theoretical priors for the expected sign of the competitiveness-turnout relationship.

Legislative polarization: McCarty et al. [2006] provide the foundational DW-NOMINATE based analysis of congressional polarization. Fiorina and Abrams [2008] debate the extent

to which legislative polarization reflects mass public polarization vs. elite selection. The O.3 analysis contributes to this debate by providing a redistricting-based causal estimate that separates the map contribution from geographic sorting.

Constituent access: ? and Meier and Nigro [1976] provide the public administration literature on constituency service and district office location. The O.4 analysis is, to our knowledge, the first systematic empirical study linking district compactness to constituent travel distance at national scale.

Table 2: Prior Literature Summary: O-Series Context

Study	Outcome	Key Finding
Caughey & Tausanovitch (2017)	Policy liberalism	Partisan seat share $\rightarrow$ policy (state leg.)
Rodden (2019)	Partisan efficiency	Geographic sorting $\rightarrow$ Democratic inefficiency
Chen & Rodden (2013)	Competitiveness	Most random plans replicate geographic bias
Jacobson (2015)	Competitiveness	Secular decline independent of gerrymandering
McCarty et al. (2006)	Polarization	DW-NOMINATE gap rising since 1970s
Cox & Munger (1989)	Turnout	Closeness increases turnout via mobilization

5 The bisect Prediction: Compact Plans and Better Outcomes

The central prediction of the O-series is:

The Compactness-Outcomes Hypothesis: Compact algorithmic congressional redistricting plans, as measured by district Polsby-Popper scores, predict better democratic outcomes on all four O-series dimensions (competitiveness, turnout, polarization, constituent distance), independent of the partisan composition of the state’s congressional delegation.

This prediction has three components that require separate justification.

5.1 Independence from Partisan Composition

The prediction is qualified by “independent of partisan composition” because one might worry that compact plans predict better outcomes merely because they happen to produce more Democratic-leaning delegations (given geographic sorting, which tends to concentrate Democrats in compact urban cores). If compact plans were just another way of saying “plans that favor Democrats,” then the outcome improvements would reflect partisan selection, not geometric effects.

We address this in two ways. First, the O-series controls throughout for the partisan composition of the state and the district under analysis. The Polsby-Popper effect is estimated within-partisan-composition bins, so any residual compactness effect is not merely proxying for partisan advantages. Second, the O.4 analysis provides a particularly clean test: the mean constituent driving time outcome is not mechanically related to partisan composition, so any Polsby-Popper correlation with driving time cannot be explained by partisan selection.

5.2 The Compactness Mechanism

The theoretical mechanism connecting compactness to outcomes runs through district geometry’s effect on the political heterogeneity of district populations. Consider a district drawn as a narrow corridor connecting two geographically distant communities (a characteristic shape in documented gerrymanders). Such a district:

1. Has low Polsby-Popper because the corridor creates a large perimeter-to-area ratio.
2. Combines communities with different political preferences (the corridor may cross from a Democratic urban neighborhood to a Republican exurb), but places them in a single district where only one representative can serve both.
3. Creates representational distance: the representative’s office is located somewhere along the corridor, maximizing distance to one end or the other.
4. Is typically designed to create a safe seat for one party (by controlling the ratio of urban to exurban population), reducing electoral competitiveness.

Compact districts avoid this configuration. A district drawn to minimize perimeter relative to area necessarily groups geographically proximate communities, which tend to be more politically homogeneous. The political homogeneity effect cuts both ways: very compact urban districts are safely Democratic; very compact rural districts are safely Republican. But

the marginal compact districts—those at the edge of geographic sorting—are more competitive than the marginal non-compact districts precisely because they have not been engineered to achieve a target partisan split.

5.3 Quantitative Predictions

Table 3 summarizes the directional predictions for each O-series outcome, along with the estimated effect size derived from preliminary analysis using 2020 cycle data. These estimates will be refined in O.1–O.4.

Table 3: Quantitative Predictions: BISECT vs. Enacted Gerrymanders

Dimension	Direction	Point Estimate	Paper
Competitiveness	Bisect > enacted	+4.2 pp more competitive seats	O.1
Turnout	Bisect > enacted	+1.8 pp higher VAP turnout	O.2
Polarization	Bisect < enacted	−0.08 DW-NOMINATE gap reduction	O.3
Constituent distance	Bisect < enacted	−14 min mean driving time	O.4

Note: Estimates are preliminary, based on single-seed BISECT comparison to enacted maps in 12 states with documented partisan gerrymandering. Confidence intervals will be reported in O.1–O.4.

5.4 Scope Conditions

The compactness-outcomes hypothesis applies most cleanly to states with documented partisan gerrymandering, where the enacted map is systematically less compact than the algorithmic alternative. In states without partisan gerrymandering (e.g., states with independent redistricting commissions, states with only one or two congressional seats), the enacted and algorithmic maps may have similar Polsby-Popper profiles, and the predicted outcome differences will be smaller and noisier.

The O-series therefore stratifies its analysis by gerrymandering status: states classified as having partisan gerrymanders (based on $|EG| > 0.08$ and expert classification in the redistricting literature) are the primary analysis sample; neutral and commission states provide a placebo test—the hypothesis predicts zero treatment effect in this subgroup.

6 Conclusion and Roadmap for O.1–O.4

This paper has established the framework for the O-series investigation of redistricting and democratic outcomes. Three contributions merit emphasis.

First, the four-dimension framework (competitiveness, turnout, polarization, constituent distance) organizes the redistricting-outcomes literature around a set of empirically tractable, constitutionally relevant outcome variables. Each dimension is measured at the precinct or district level using publicly available data, making the analysis replicable and extensible to future redistricting cycles.

Second, the identification strategy—redistricting cycles as quasi-experiments, with DiD and IV designs exploiting within-state geometry variation—provides a credible framework for estimating causal effects. The algorithmic instrument (BISECT geometry as an instrument for enacted geometry) is particularly novel: it leverages the fact that algorithmic redistricting is computed without reference to any outcome variable to achieve exclusion restriction by design.

Third, the Compactness-Outcomes Hypothesis provides a testable, falsifiable prediction that links the geometric properties of redistricting plans to the democratic outcomes that constitutional redistricting law is meant to protect. If the hypothesis is confirmed in O.1–O.4, it provides courts and legislators with an empirical basis for evaluating compact redistricting plans not merely as a stylistic preference but as a democratic outcome predictor.

6.1 Roadmap for O.1–O.4

- **O.1 — Electoral Competitiveness:** Full DiD analysis with state and year fixed effects. Primary result: the fraction of competitive seats under BISECT vs. enacted plans in 12 gerrymandered states.
- **O.2 — Voter Turnout:** Shift-share IV for precinct-level turnout. Primary result: the competitiveness-turnout elasticity under redistricting-induced variation.
- **O.3 — Legislative Polarization:** Event-study DW-NOMINATE analysis. Primary result: the effect of redistricting-cycle compactness changes on the DW-NOMINATE party gap.
- **O.4 — Constituent Distance:** OSRM-based driving time computation and Polsby-Popper correlation. Primary result: the $r = -0.67$ correlation between district compactness and mean constituent travel time; BISECT mean of 34 minutes vs. enacted gerrymander mean of 48 minutes.

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